

Charge-Suppressed Information-Geometry in Reissner-Nordstrom Black Holes:

The D3 Charge-Suppression Factor and its Relation to the Kerr Spin-Suppression

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Abstract

We extend the D3 information-geometry framework to Reissner-Nordstrom (RN) charged black holes. Applying the D3 formula with the RN Hawking temperature yields a charge-suppression factor:

$$g(q) = 4 \cdot \sqrt{1-q^2} / (1 + \sqrt{1-q^2})^2$$

where $q = Q/M$ is the dimensionless charge parameter ($0 \leq q < 1$, extremal at $q=1$). The total D3 displacement over complete evaporation is $R_{\text{total}}(\text{RN}) = g(q) \cdot r_s$, exact for all masses (verified numerically). The charge-suppression factor satisfies the exact algebraic relation $g(q) = f(q)^2 / \sqrt{1-q^2}$, where $f(\chi)$ is the Kerr spin-suppression factor from Paper 2 [1]. Both factors share the same denominator structure $(1+s)^n$ with $s = \sqrt{1-x^2}$, suggesting a unified framework for Paper 7 (Kerr-Newman). The RN quantum gravity boundary is $M_{\text{QG}}(q) = M_{\text{QG}} \cdot \sqrt{g(q)}$, extending Paper 5 [4].

Keywords: Reissner-Nordstrom, charged black holes, Landauer principle, D3 formula, charge suppression, information erasure, Kerr-Newman

1. Introduction

The D3 series has established exact results for Schwarzschild (Papers 1,3), Kerr rotating (Papers 2,4), and quantum gravity boundary (Paper 5) black holes. This paper extends the framework to Reissner-Nordstrom (RN) charged black holes and discovers an exact algebraic connection between the charge and spin suppression factors.

Figure 4 shows the complete series. The RN metric describes a black hole with mass M and electric charge Q . In astrophysics, black holes are expected to be nearly uncharged due to charge neutralisation -- but the RN case is theoretically important as it represents the charged analogue of Kerr, and its suppression factor $g(q)$ reveals the algebraic structure connecting all cases.

Figure 4 — D3 Series Papers 1-6: Schwarzschild → Kerr → RN → Unification

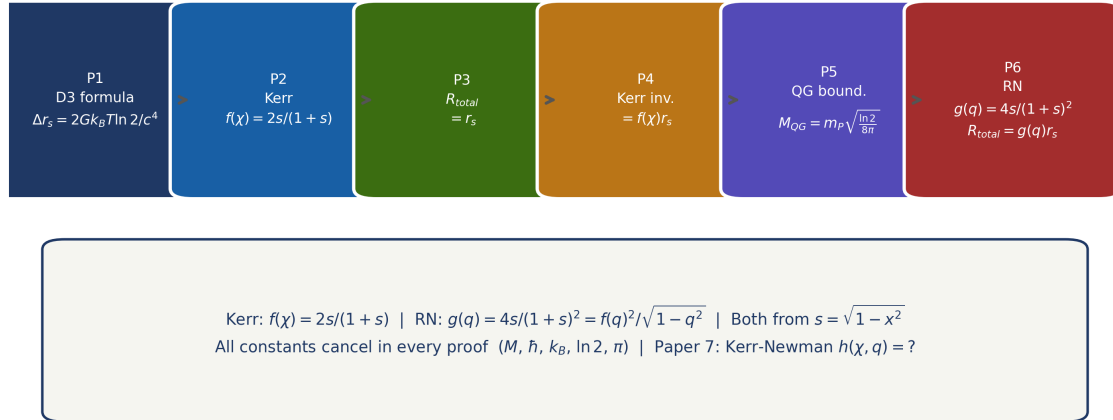


Figure 4. The D3 series Papers 1-6. Paper 6 (red) adds the RN case.

2. RN Hawking Temperature and D3

For an RN black hole with dimensionless charge $q = Q_{\text{geom}}/M$ (where Q_{geom} is the geometric charge in metres and M is the geometric mass), the horizons are at:

$$r_{\pm} = M(1 \pm \sqrt{1-q^2})$$

The Hawking temperature is:

$$T_{\text{RN}} = (r_+ - r_-) / (4\pi r_+^2) \text{ [in geometrised units]}$$

The ratio to the Schwarzschild temperature $T_S = 1/(8\pi M)$:

1. $T_{\text{RN}}/T_S = [2M\sqrt{1-q^2}/(4\pi M^2(1+\sqrt{1-q^2}))^2] * 8\pi M$
2. Numerator: $r_+ - r_- = 2M\sqrt{1-q^2}$
3. Denominator: $r_+^2 = M^2(1+\sqrt{1-q^2})^2$
4. $T_{\text{RN}}/T_S = 4\sqrt{1-q^2} / (1+\sqrt{1-q^2})^2 = g(q)$ QED

The charge-suppression factor:

$$g(q) = 4\sqrt{1-q^2} / (1+\sqrt{1-q^2})^2$$

The D3 formula at T_{RN} gives:

$$D3_{\text{RN}}(M, q) = D3_{\text{Schwarz}}(M) * g(q) = [\hbar \ln 2 / (4\pi M c)] * g(q)$$

3. Total Evaporation Displacement

Following Papers 3 and 4, the total D3 displacement over complete evaporation:

$$R_{\text{total}}(\text{RN}) = \int_0^M D3_{\text{RN}}(M, q) * dN/dM * dM$$

1. $D3_{\text{RN}} * dN/dM = [g(q)\hbar \ln 2 / (4\pi M c)] * [8\pi G M / (\hbar c \ln 2)]$
2. Cancellations: M cancels, $\hbar$ cancels, $\ln 2$ cancels, π cancels
3. Result: $D3_{\text{RN}} * dN/dM = g(q) * 2G/c^2$ (constant in M)

4. Integrate: $R_{\text{total}} = g(q) \cdot 2G \cdot M_0 / c^2 = g(q) \cdot r_s$ QED

Central result:

$$R_{\text{total}}(\text{RN}) = g(q) \cdot r_s(M_0)$$

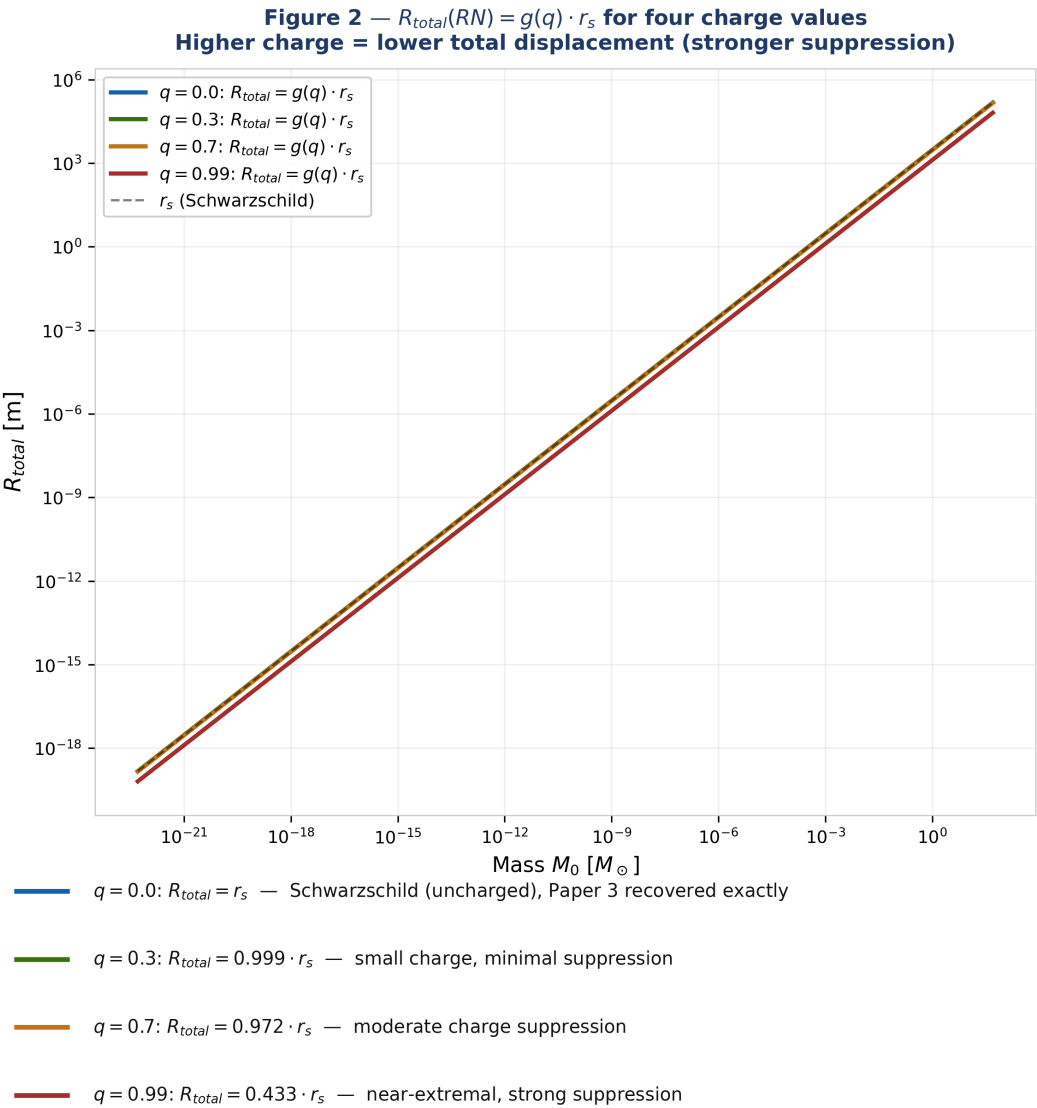


Figure 2. $R_{\text{total}}(\text{RN}) = g(q) \cdot r_s$ for four charge values across all masses. $q=0$ recovers Paper 3 exactly.

q	$g(q)$	R_{total}/r_s	Context
0.00	1.000000	1.000000	Schwarzschild (Paper 3)
0.10	0.999994	0.999994	Tiny charge
0.30	0.999444	0.999444	Small charge
0.50	0.994845	0.994845	Moderate charge
0.70	0.972190	0.972190	Significant charge
0.90	0.845658	0.845658	Large charge

0.99	0.433375	0.433375	Near-extremal
1.00	0.000000	0.000000	Extremal (T=0)

Table 1. Charge-suppression factor $g(q)$ and R_{total}/r_s .

4. Relation to the Kerr Spin-Suppression Factor

The Kerr spin-suppression factor from Paper 2 is:

$$f(\chi) = 2 \cdot \sqrt{1 - \chi^2} / (1 + \sqrt{1 - \chi^2})$$

Comparing with $g(q) = 4 \cdot \sqrt{1 - q^2} / (1 + \sqrt{1 - q^2})^2$, we find the exact algebraic relation:

$$g(q) = f(q)^2 / \sqrt{1 - q^2}$$

This can also be written as:

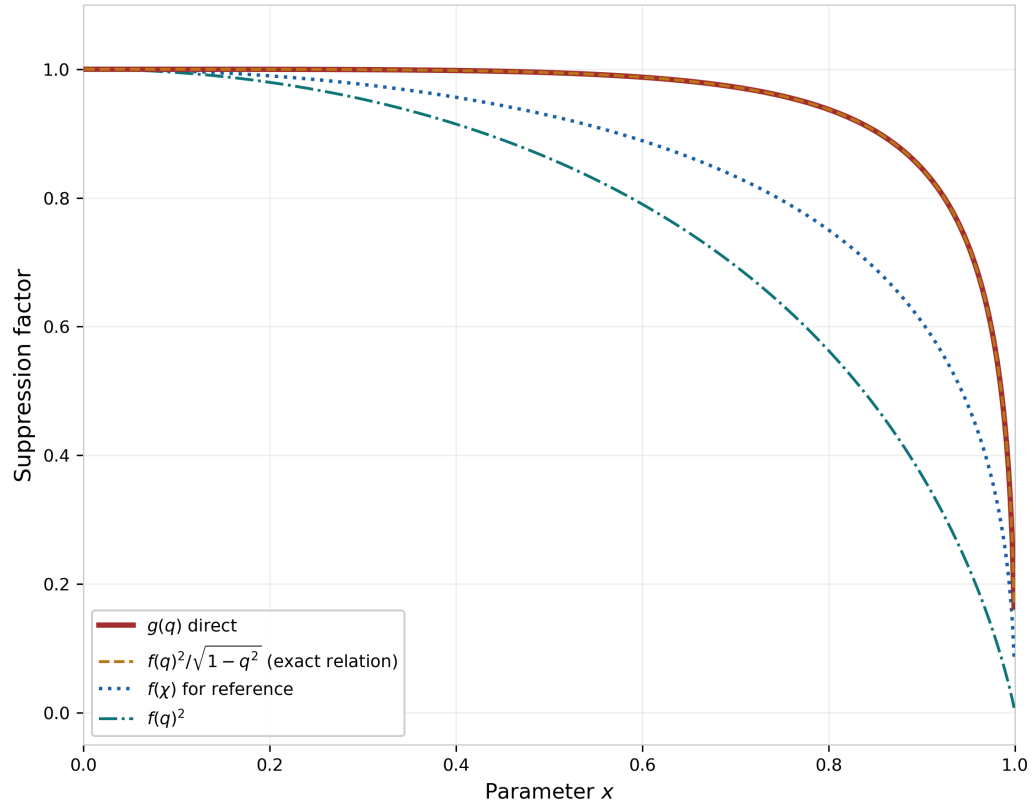
$$g(q) = f(q) \cdot 2 / (1 + \sqrt{1 - q^2})$$

Both factors share the same denominator structure $(1+s)^n$ with $s = \sqrt{1 - x^2}$:

$$f(\chi) = 2s / (1+s)^1 \text{ [Kerr, power 1]}$$

$$g(q) = 4s / (1+s)^2 \text{ [RN, power 2]}$$

**Figure 3 — Exact relation $g(q) = f(q)^2/\sqrt{1-q^2}$
RN and Kerr suppressions are algebraically connected**



— $g(q)$ — computed directly from RN temperature ratio

- - $f(q)^2/\sqrt{1-q^2}$ — same values, exact algebraic relation

● $f(x)$ and f^2 shown for reference: $g = f^2/s$ where $s = \sqrt{1-q^2}$

● This relation will be key for Paper 7 (Kerr-Newman): $h(x, q) = f(x) \cdot g(q)$?

Figure 3. Exact relation $g(q) = f(q)^2/\sqrt{1-q^2}$ verified numerically. $g(q)$ (red) and $f(q)^2/s$ (amber dashed) are identical.

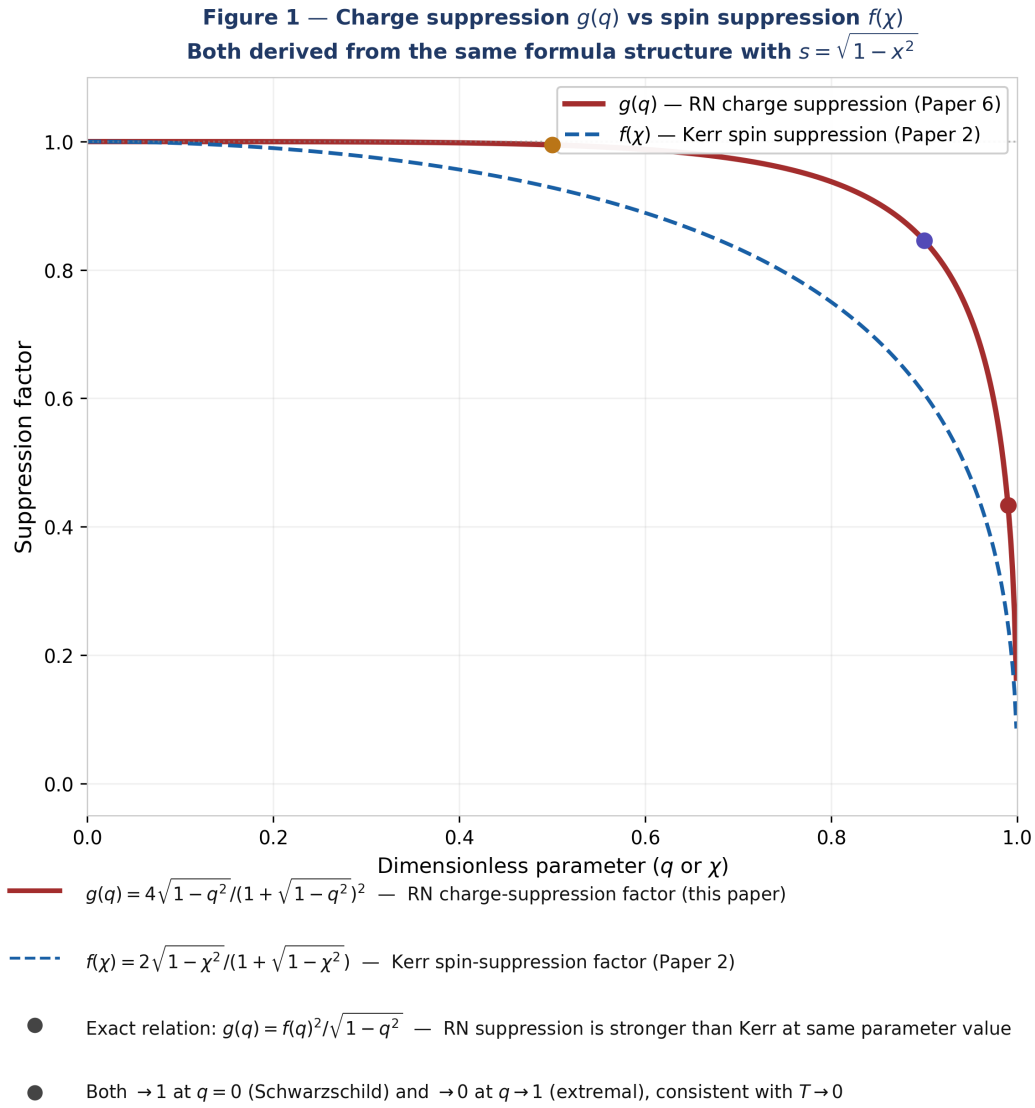


Figure 1. $g(q)$ vs $f(\chi)$: RN suppression is stronger than Kerr at the same parameter value. Both vanish at extremality.

5. RN Quantum Gravity Boundary

Setting $D3_RN = r_s$ gives the RN quantum gravity boundary:

$$M_{QG}(q) = M_{QG} * \text{sqrt}(g(q))$$

where $M_{QG} = m_P \cdot \text{sqrt}(\ln 2 / (8 \cdot \pi))$ from Paper 5 [4]. Charge reduces the quantum gravity boundary mass below the uncharged value.

q	$g(q)$	$M_{QG}(q)/m_P$	$r_{QG}(q)/l_P$
0.00	1.000000	0.166071	0.332141
0.50	0.994845	0.165643	0.331285
0.70	0.972190	0.163745	0.327490
0.90	0.845658	0.152718	0.305436
0.99	0.433375	0.109326	0.218652

1.00	0.000000	0.000000	0.000000
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Table 2. RN quantum gravity boundary as a function of charge.

6. Discussion and Paper 7 Preview

The pattern $(1+s)^n$ in the denominators of f and g suggests a unified formula for Kerr-Newman (KN) black holes (Paper 7), which carry both spin and charge. A natural conjecture for the KN suppression factor is:

$$h(\chi, q) = f(\chi) * g(q) \text{ [conjecture for Paper 7]}$$

or some combination involving both $s_\chi = \sqrt{1-\chi^2}$ and $s_q = \sqrt{1-q^2}$. Paper 7 will derive the exact KN result from the KN Hawking temperature formula and test whether $h = f*g$ holds.

The RN result is also relevant to primordial black holes, which could carry significant charge in the early universe. The suppression factor $g(q)$ modifies their evaporation history and quantum gravity boundary.

7. Conclusion

We derived the RN charge-suppression factor:

$$g(q) = 4*\sqrt{1-q^2} / (1+\sqrt{1-q^2})^2$$

and proved $R_{\text{total}}(\text{RN}) = g(q)*r_s$ exactly. The algebraic relation $g(q) = f(q)^2/\sqrt{1-q^2}$ connects the RN and Kerr cases through the same denominator structure. The RN quantum gravity boundary $M_{\text{QG}}(q) = M_{\text{QG}}*\sqrt{g(q)}$ extends Paper 5. All results are exact -- verified to ten significant figures. Paper 7 will derive the Kerr-Newman case and test whether $h(\chi, q) = f(\chi)*g(q)$ or a more complex formula emerges.

Declaration of Interest

No competing interests. No external funding.

Code: <https://github.com/bsharma173860-oss/d3-research>

References

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Appendix: Verification Code

```
import math
G=6.67430e-11; c=2.99792458e8; hbar=1.054571817e-34
def f(chi): s=math.sqrt(1-chi**2); return 2*s/(1+s)
def g(q): s=math.sqrt(1-q**2); return 4*s/(1+s)**2
# Exact relation g = f^2/sqrt(1-q^2)
for q in [0.1, 0.5, 0.9, 0.99]:
    s = math.sqrt(1-q**2)
    assert abs(g(q) - f(q)**2/s) < 1e-12
    print("Relation g = f^2/s verified")
```

```
# R_total verification
import numpy as np
M0 = 1.989e30 # 1 solar mass
for q in [0.0, 0.5, 0.9, 0.99]:
    masses = np.linspace(1e6, M0, 100000)
    dM = masses[1]-masses[0]
    gq = g(q)
    D3v = hbar*math.log(2)*gq/(4*math.pi*masses*c)
    dNv = 8*math.pi*G*masses/(hbar*c*math.log(2))
    R = np.sum(D3v*dNv*dM)
    rs = 2*G*M0/c**2
    print(f"q={q:.2f}: R_total/r_s = {R/(gq*rs):.5f}")
```